

**NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY, EAST CAMPUS, GEETA COLONY
NEW DELHI**

Total no. of Pages--01

Roll no.

THIRD SEMESTER – B.TECH

MID-SEMESTER EXAMINATION, September 2025

Course Code- EAEPC302

Course Title- Signal and Systems

Time- 1:30 Hours

Max. Marks- 20

Note: - Attempt all questions. Missing data/ information if any, may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Determine whether the following system is: $y[n] = x[n]x[n-2]$ i) Causal ii) Linear	2	CO1, 2
1b	Determine whether the following system is: i) Memoryless, ii) Time-invariant. $y(t) = t \exp(x(t))$	2	CO1, 2
2a	$x[n] = \begin{cases} 1, & n = 1 \\ -1, & n = -1 \\ 0, & n = 0 \text{ and } n > 1 \end{cases}$ For given $x[n]$, find $y[n] = x[n] + x[-n]$ and sketch $y[n+2]$ and $y[n/2]$.	2	CO 1, 2, 3
2b	Show that the product of two even signals or of two odd signals is an even signal and that the product of an even and an odd signal is an odd signal.	2	CO 2, 5
3a	For $x[n] = \{1, 2\}$ at $n=0, 1$ and $h[n] = \{1, -1, 1\}$ at $n=0, 1, 2$ compute $y[n] = x[n] * h[n]$.	2	CO 2, 5
3b	Let $x[n] = 3\delta[n] + \delta[n-1]$ and $h[n] = 2\delta[n] + \delta[n-1]$ Compute $y[n] = x[n] * h[n]$	2	CO 2
4a	Determine the discrete-time convolution sum of the given sequences $x[n] = [1, 2, 3, 4]$ and $h[n] = [1, 5, 1]$	2	CO 2
4b	Determine the condition of the impulse response of the LTI system if system is, i) Memory less ii) Stable	2	CO 4, 5
5a	Find the convolution integral of $x(t)$ and $h(t)$, $x(t) = (t-1)\{u(t-1) + u(t-3)\}$ and $h(t) = [u(t+1) - 2u(t-2)]$.	2	CO 2, 4
5b	Sketch the convolved signal of Prob. 5a.	2	CO 2, 4